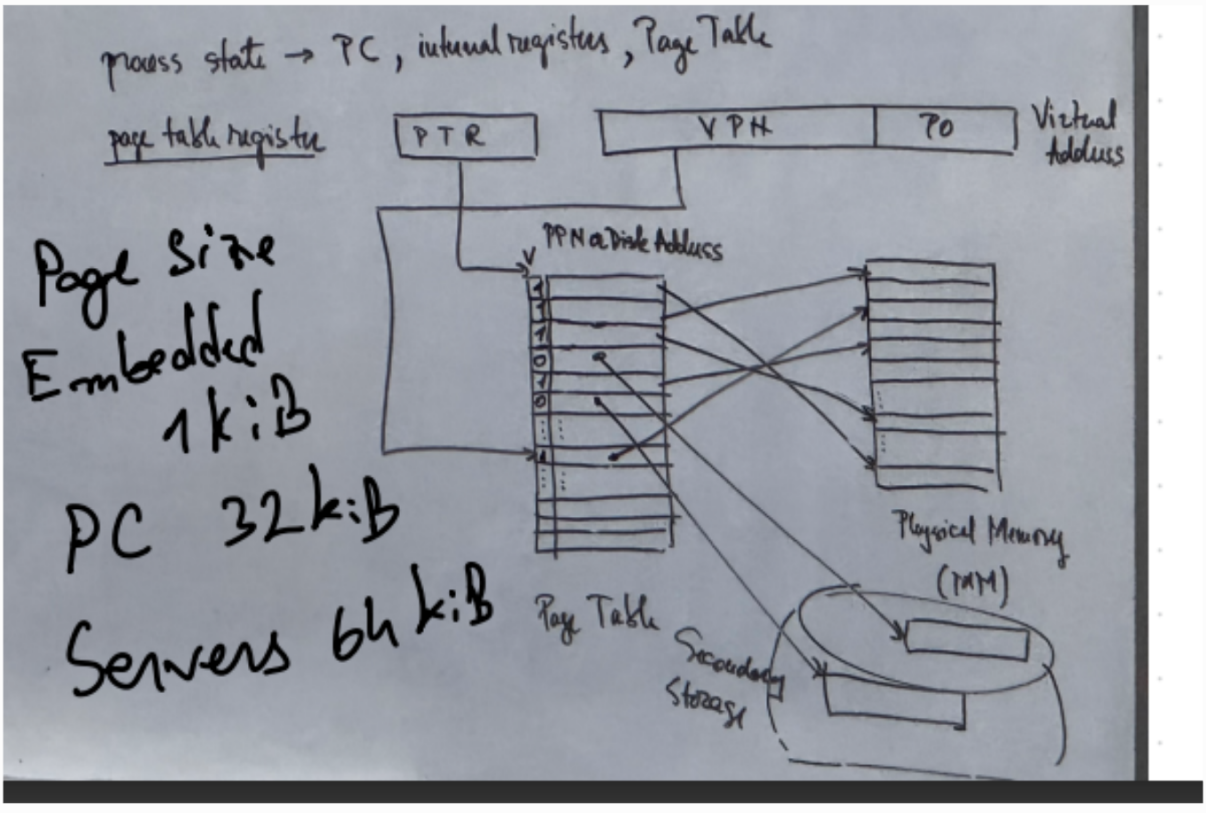
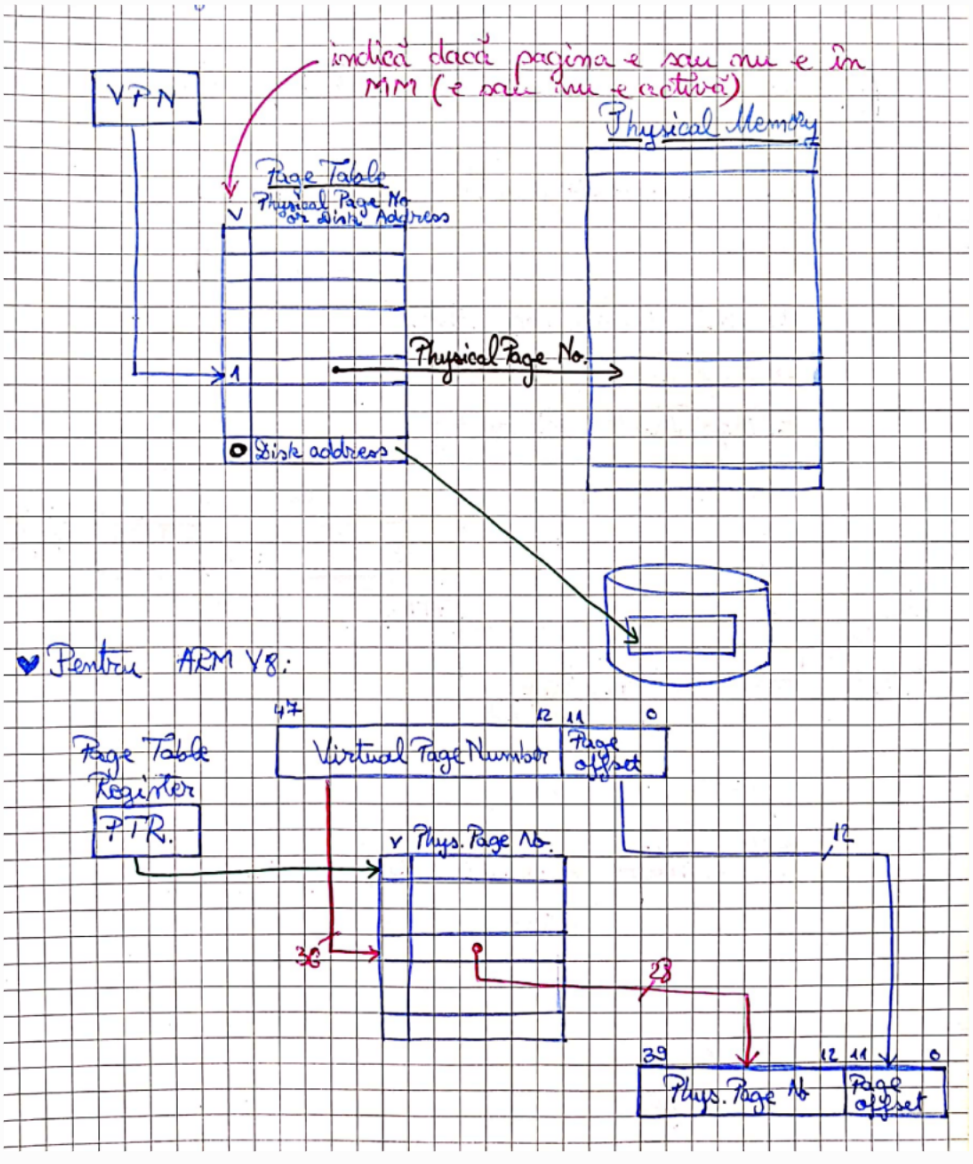
4. 2. Page Placement



V=1 => ∃ MM

V=0 => ∄ SS



Problems: (1) Page Table Size

ARM - 1 PT Entry = 64 bits = 2^3 B

$2^{36} \times 2^3 = 2^{39} = 0{,}5$ TiB

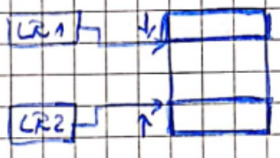
(2) Page Table Reads (Page Table is placed in the Physical Memory)

Solutions:

(1) A. Limit register



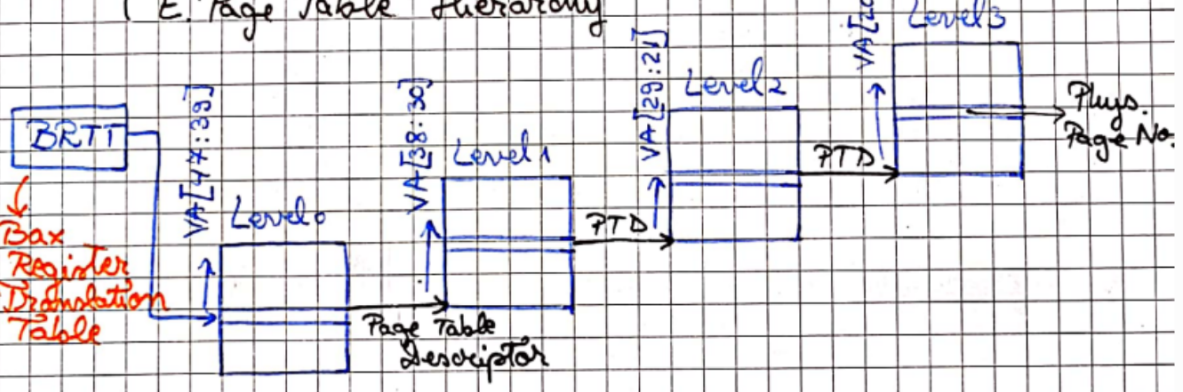
B. Heap & Stack growth



C. Hash (inverse page table)

D. Page the Page Table

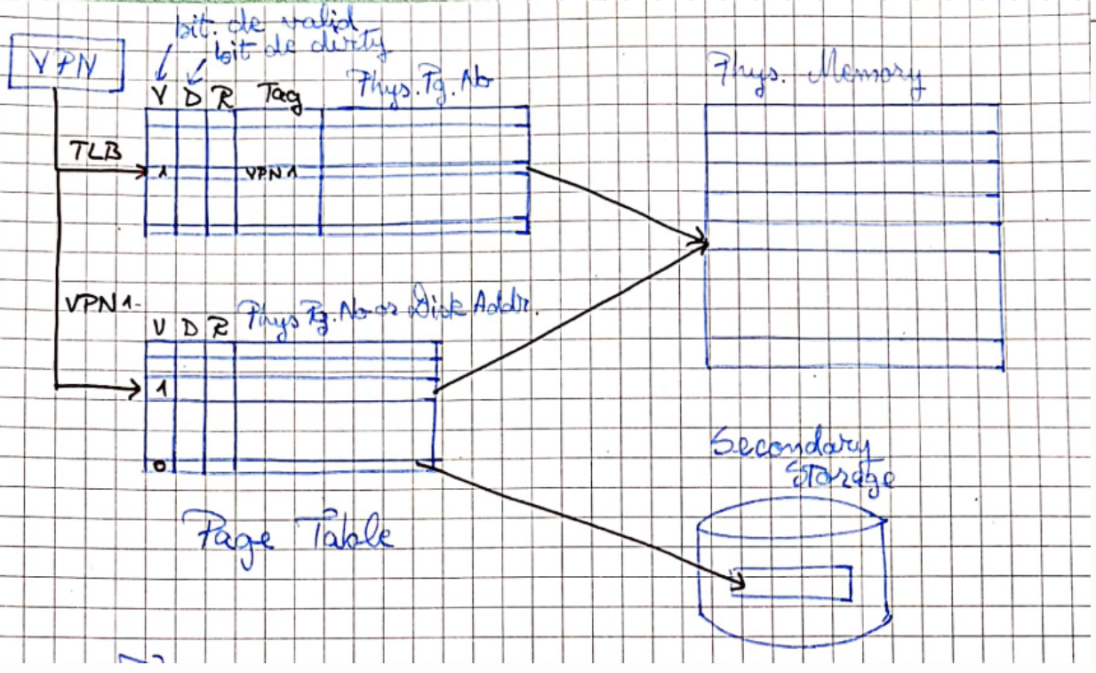
E. Page Table Hierarchy

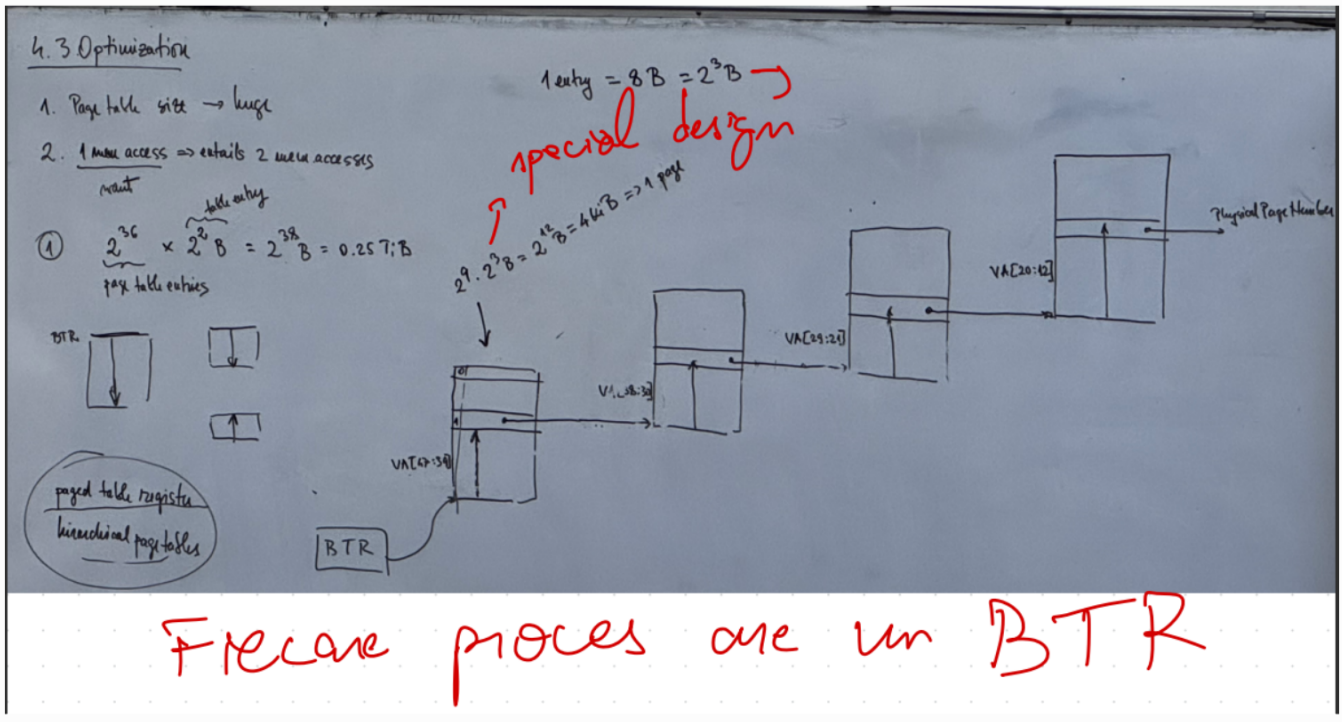


$2^9 \times 2^3$ B $= 2^{12}$ B $= 4$ KiB

(2) Translation Lookaside Buffer (TLB)

- Cache the Page Table

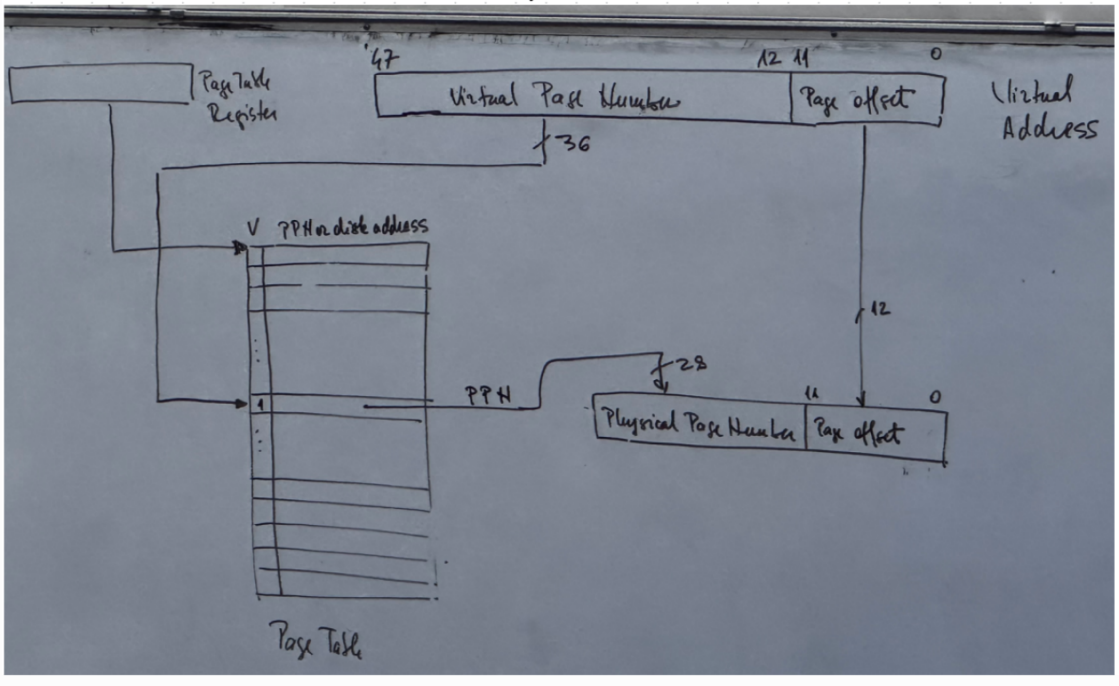




Fiecare proces are un BTR

② Cached Page Tables → Translation Lookaside Buffer (TLB)

-> small size (FA, SA, SM)
-> block size -> 1-2 page table entries



Intensity's FAST MATH

TLB 16-512 entries

block size = 1-2 table entries

hit time = 1 CC

miss penalty 10s-100s cc

miss rate 0,1-1%

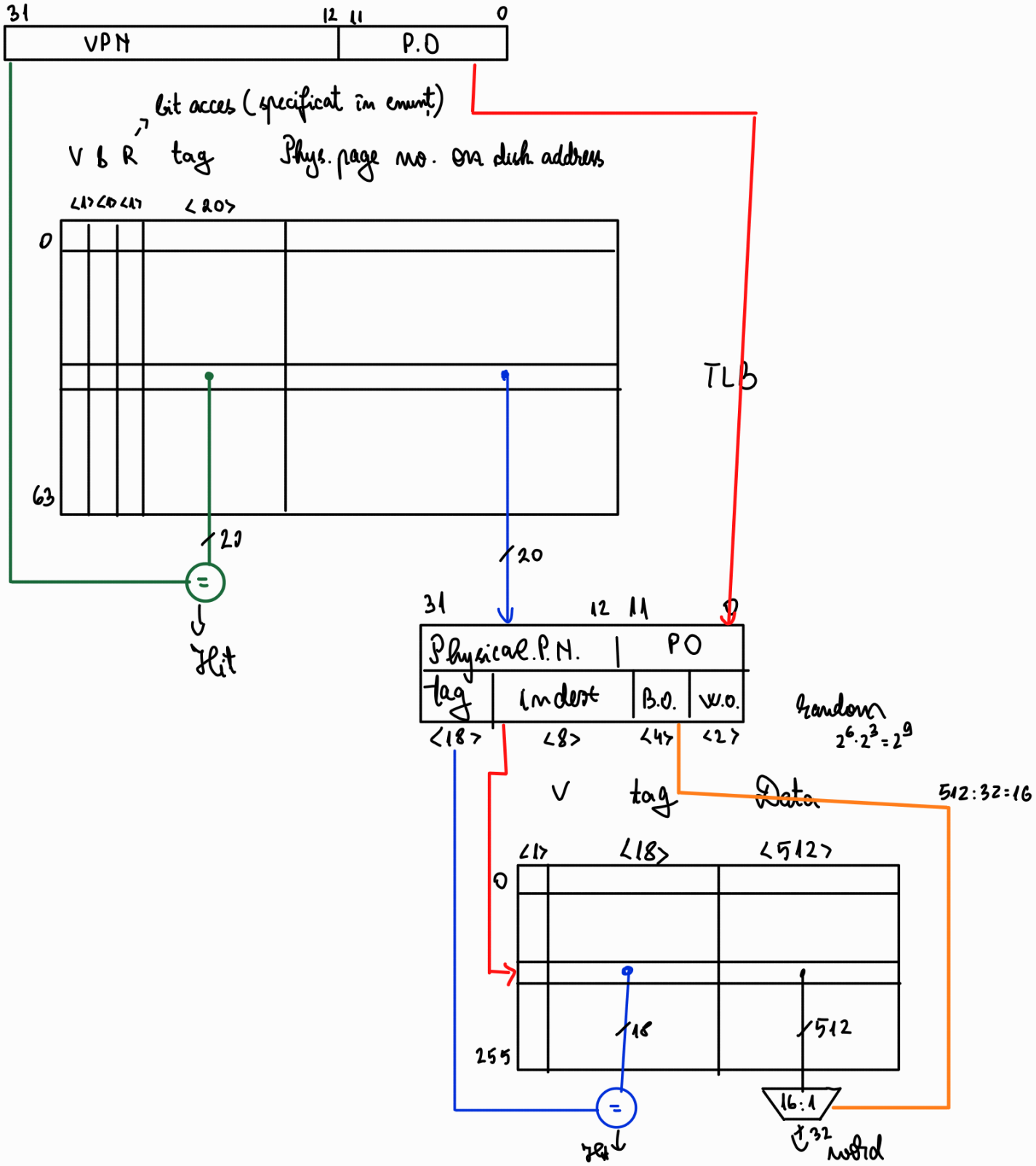
virtual address = 32 b = 4 GiB

physical address = 32 b = 4 GiB

page size = 4 KiB = $2^2 \cdot 2^{10} = 2^{12}$ P.O.

1 word = 4 B

1 block = 16 words = 2^4



(2.) VM → 64 TiB

PM → 8 GiB

byte addressing

16 KiB pages

32-entries, 8M TLB

1 block = 2 page table

entries = 2^1

256 KiB cache size (2 way SA)

1 block = 16 words

1 word = 64 bits

---

1 word = 64 bits = 8 B = 2^3 B ⇒ W.O. = 3 b

1 block = 16 words = 2^4 words ⇒ B.O. = 4 b

256 KiB cache = $2^8 \cdot 2^{10}$ B = 2^{18} B ⇒ $\frac{2^{18}}{2^4 \cdot 2^3 \cdot \underset{\uparrow}{2^1}} = 2^{10}$ ⇒ index = 10 b
2SA

P.M. = 8 GiB = $2^3 \cdot 2^{30} = 2^{33}$ ⇒ P.M. address = 33 b

VM. = 64 TiB = $2^6 \cdot 2^{40} = 2^{46}$ ⇒ V.M. address = 46 b

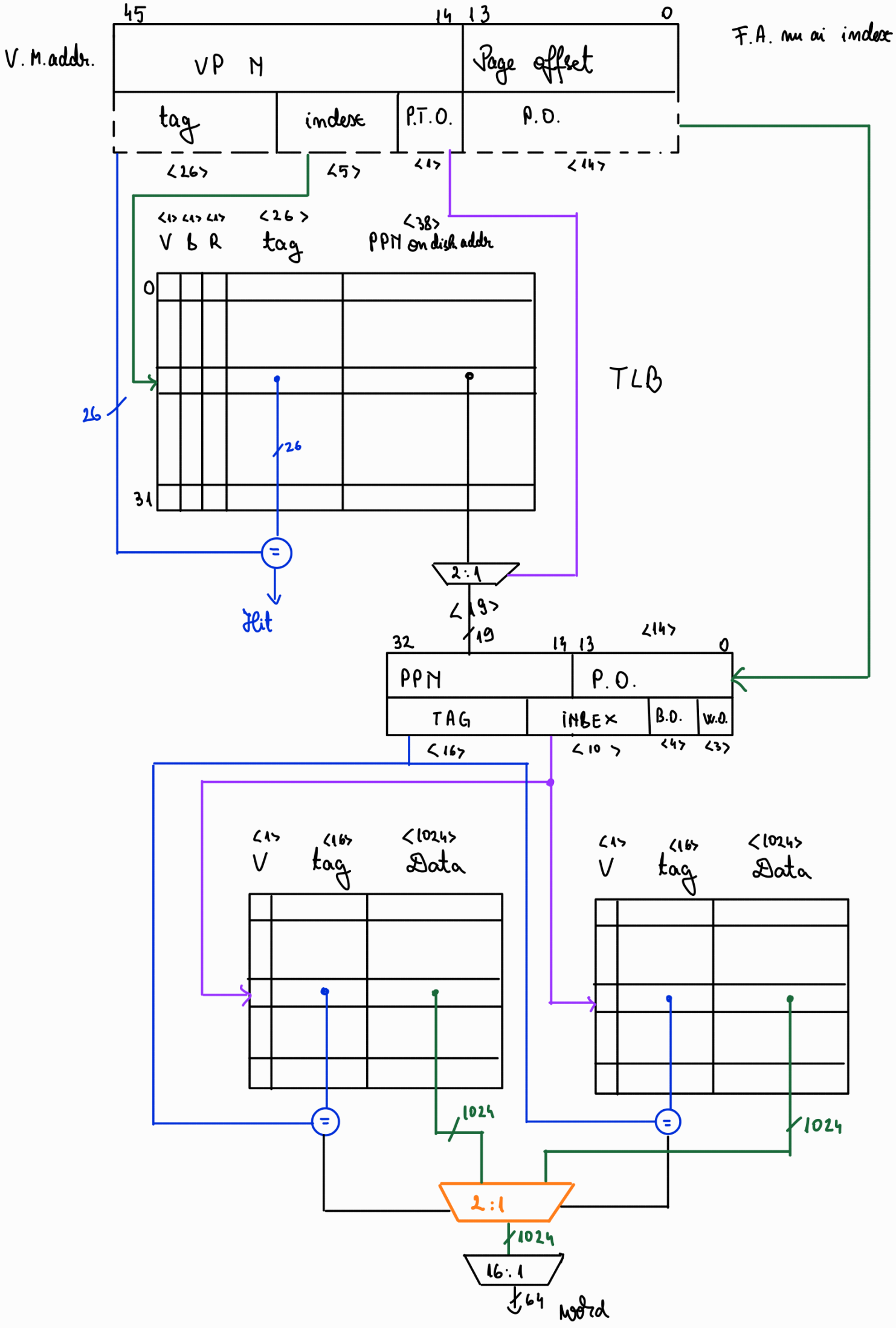
16 KiB pages = $2^4 \cdot 2^{10}$ pages = 2^{14} pages ⇒ P.O. = 14 b

1 block = 2 page table = 2^1 ⇒ P.T. offset = 1 b

32 entries = 2^5 ⇒ index = 5 b

⇒ tag = 46 - 14 - 1 - 5 = 26 b

} ⇒ tag = 33 - 17 - 16 b


V. M. addr.
45
14 13
0
VP N
Page offset
tag
index
P.T.O.
P.O.
F.A. nu ai index
<26>
<5>
<1>
<14>
<1> <1> <1>
V b R
<26>
tag
<38>
PPN on disk addr
0
31
26
26
=
Hit
TLB
2:1
<19>
19
32
14 13
<14>
0
PPN
P.O.
TAG
INDEX
B.O.
W.O.
<16>
<10>
<4>
<3>
<1>
V
<16>
tag
<1024>
Data
<1>
V
<16>
tag
<1024>
Data
=
=
1024
1024
2:1
1024
16:1
64
word